AIMS DTU Summer Project 2026

Digital Twin of a Scientist

Preview

Build an AI agent that emulates a famous scientist or researcher, reproducing not just what they knew but how they reasoned, taught, and spoke.

Most chatbots answer in a generic voice. A Digital Twin is different: it should feel like a specific person. The goal is an agent that accurately emulates one chosen scientist's knowledge, reasoning style, communication style, and research expertise, so that talking to it feels like talking to them. A good twin gets the facts right, frames problems the way that person would, and stays in character across a long conversation.

This rests on three pillars. A retrieval-augmented generation (RAG) pipeline grounds answers in the scientist's actual work, so the agent cites real ideas instead of inventing them. A memory system gives the agent both short-term recall within a conversation and long-term recall across sessions. A carefully designed persona keeps the voice, values, and teaching style consistent throughout.

Your Assignment

Pick one scientist or researcher and build their Digital Twin. The work has the following components:

- Choose one scientist or researcher with enough public material to draw from, such as papers, books, interviews, and lectures.
- Implement proper memory, covering both conversation memory within a session and long-term memory that persists across sessions.
- Build a RAG pipeline over the scientist's papers, articles, books, interviews, lectures, and similar sources.
- Answer advanced questions in the scientist's domain accurately and in depth.
- Maintain consistency with the scientist's personality and teaching style throughout the conversation.
- Support multi-turn conversations while remembering previous context.

Use Gemini 2.5 Flash, so that evaluation remains consistent across submissions and API endpoints are easily available.

Evaluation

Your Digital Twin will be assessed on:

- Persona consistency, meaning how reliably it sounds and reasons like the scientist.
- Technical accuracy of its answers within the scientist's domain.
- Memory quality, across both short-term and long-term recall.
- RAG quality, meaning how well answers are grounded in retrieved sources.
- Overall user experience of holding a conversation with the twin.

Deliverables

1. A GitHub repository with clean, reproducible code for the agent, RAG pipeline, and memory system.
2. A working demo that can be run interactively.
3. An architecture diagram showing how the components fit together.
4. Ten sample conversations that showcase persona, accuracy, and memory.
5. Short documentation explaining your approach and design decisions.

Bonus

Extra credit for going beyond the core requirements:

- Voice interaction for speaking to and hearing from the twin.
- A memory visualization dashboard showing what the agent remembers.
- Make the agent timeline aware.

Pick your scientist and start building.